

human

Show me homes in Exeter

host_agent

Calling tool send_message with args : {'target': 'planning_agent', 'task_description': 'Show me homes in Exeter. Provide the general steps to solve this query.'}

planning_agent

The area is Exeter, so there is 1 entry for Exeter. The entities to search for are homes, and I need to find as many entries as possible for homes in Exeter.
output steps: ["Find Exeter 1 area", "Search for homes in Exeter as many search results"]

host_agent

Tool send_message returned result : The area is Exeter, so there is 1 entry for Exeter. The entities to search for are homes, and I need to find as many entries as possible for homes in Exeter.
output steps: ["Find Exeter 1 area", "Search for homes in Exeter as many search results"]

Calling tool send_message with args : {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'}

Tool call_os_ngd returned result : ['Search results have been found. \n Artifact generated is : exeter_area_polygon,\n Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter..,\n count of records fetched : 98.\n ',]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['exeter_area_polygon']}

Tool generate_metadata_for_artifacts returned result : ...

Tool Code Executor

```
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where any of the name columns contain 'Exeter' (case-insensitive)
    mask = (
        gdf['name1_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name2_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name3_text'].str.contains('Exeter', case=False, na=False) |
        gdf['name4_text'].str.contains('Exeter', case=False, na=False)
    )
    filtered = gdf[mask]
    summary = f"Filtered polygons for Exeter. Number of matching polygons: {len(filtered)}."
    artifact_name = 'filtered_exeter_area_polygon'
    artifact_description = 'A GeoDataFrame containing only the polygons whose name columns contain the word Exeter.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered polygons for Exeter. Number of matching polygons: 1.',]

Filtered polygons for Exeter. Number of matching polygons: 1. An artifact has been generated with the name: filtered_exeter_area_polygon and description: A GeoDataFrame containing only the polygons whose name columns contain the word Exeter.

named_area_agent

The specific area polygon for Exeter has been filtered and saved as filtered_exeter_area_polygon.

host_agent

Calling tool send_message with args : {'target': 'buildings', 'task_description': 'Search for homes (residential buildings) within the filtered_exeter_area_polygon.'}

buildings_agent

Calling tool call_os_ngd with args : {'filters': None, 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'exeter_residential_buildings'}

Tool call_os_ngd returned result : ["Multiple search results have been found in multiple datasets. A summary of each is provided.\n Artifacts generated are : ['buildings_exeter_residential_buildings', 'buildingline_exeter_residential_buildings', 'buildingpart_exeter_residential_buildings'],\n Descriptions are : ['A geopandas dataframe containing building data with filters and bbox applied for the query Search for homes (residential buildings) within the filtered_exeter_area_polygon. using filters None.', 'A geopandas dataframe containing building line data with filters and bbox applied for the query Search for homes (residential buildings) within the filtered_exeter_area_polygon. using filters None.', 'A geopandas dataframe containing building part data with filters and bbox applied for the query Search for homes (residential buildings) within the filtered_exeter_area_polygon. using filters None.'],\n counts of records fetched are : [69169, 69342, 85480].\n ", [, ,]]

Calling tool send_message with args : {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'buildings_exeter_residential_buildings' to find homes where buildinguse_addresscount_residential > 0 and buildinguse_addresscount_total = 1. Save the filtered results as 'filtered_homes_exeter'. Provide a summary of the findings."}

data_analysis_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['buildings_exeter_residential_buildings']}

Tool generate_metadata_for_artifacts returned result : ...

Tool Code Executor

```
def filter_homes_exeter(data: list):
    gdf = data[0]
    filtered = gdf[(gdf['buildinguse_addresscount_residential'] > 0) & (gdf['buildinguse_addresscount_total'] == 1)]
    summary = f"Filtered {len(filtered)} homes in Exeter where there is exactly one address and it is residential."
    artifact_name = 'filtered_homes_exeter'
    artifact_description = 'GeoDataFrame of homes in Exeter with exactly one address, which is residential.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code_executor returned result : ['Filtered 38848 homes in Exeter where there is exactly one address and it is residential.',]

Filtered 38,848 homes in Exeter where there is exactly one address and it is residential. The results are saved as 'filtered_homes_exeter', which contains a GeoDataFrame of these homes.

buildings_agent

I found 38,848 homes in the Exeter area where there is exactly one residential address. The filtered results are saved as 'filtered_homes_exeter'.

host_agent

Calling tool send_message with args : {'target': 'plotting_agent', 'task_description': "Plot the homes from the artifact 'filtered_homes_exeter' within the area of Exeter from 'filtered_exeter_area_polygon'."}

plotting_agent

Calling tool generate_metadata_for_artifacts with args : {'artifact_names': ['filtered_homes_exeter', 'filtered_exeter_area_polygon']}

Tool generate_metadata_for_artifacts returned result : ...

Tool Code Executor

```
def plot_homes_within_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    homes_gdf = data[0].copy()
    exeter_gdf = data[1].copy()

    # Ensure CRS is set and convert to EPSG:4326 for folium
    if homes_gdf.crs is None:
        homes_gdf.set_crs(epsg=27700, inplace=True) # Assuming British National Grid
    if exeter_gdf.crs is None:
        exeter_gdf.set_crs(epsg=27700, inplace=True)
    homes_gdf = homes_gdf.to_crs(epsg=4326)
    exeter_gdf = exeter_gdf.to_crs(epsg=4326)

    # Get the Exeter polygon (assuming only one row)
    exeter_poly = exeter_gdf.iloc[0].geometry

    # Filter homes within Exeter polygon
    homes_within = homes_gdf[homes_gdf.within(exeter_poly)]

    # Create folium map centered on Exeter
    centroid = exeter_poly.centroid
```

```

m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

# Add Exeter area polygon
folium.GeoJson(mapping(exeter_poly), name='Exeter Area', style_function=lambda x: {'fillColor': '#0000ff20', 'color': '#0000ff', 'weight': 2}).add_to(m)

# Add homes polygons
for _, row in homes_within.iterrows():
    folium.GeoJson(mapping(row.geometry),
                    name=row.get('description', 'Home'),
                    style_function=lambda x: {'fillColor': '#ff780020', 'color': '#ff7800', 'weight': 1},
                    tooltip=row.get('description', 'Home')).add_to(m)

# Save map
map_filename = 'homes_within_exeter_map.html'
m.save(map_filename)

# Prepare summary
summary = f"Plotted {len(homes_within)} homes within the Exeter area. First 5 homes: "
summary += str(homes_within[['osid', 'description', 'geometry_area_m2']].head(5).to_dict('records'))

return [summary, 'homes_within_exeter_map', 'Folium map showing homes within the Exeter area polygon, with Exeter boundary and homes highlighted.', map_fil

```

Tool code_executor returned result : ["Plotted 38841 homes within the Exeter area. First 5 homes: [{'osid': 'b9a49c80-fecf-4308-a98d-cf90ea4ea2e4', 'description': 'End-Of-Terrace House', 'geometry_area_m2': 99.338}, {'osid': '9dfb4b41-f53d-44c0-8a7a-5f7884375677', 'description': 'Mid-Terrace House', 'geometry_area_m2': 78.565}, {'osid': 'aea058f2-6a39-4829-ae6b-85c6863a250e', 'description': 'Mid-Terrace House', 'geometry_area_m2': 66.91}, {'osid': 'a578be0a-7fff-4fbb-8ace-2cf755bbb53c', 'description': 'End-Of-Terrace House', 'geometry_area_m2': 51.92}, {'osid': 'c86c6a23-5ced-4dae-be17-94f33e02c66d', 'description': 'Mid-Terrace House', 'geometry_area_m2': 51.91}]",]

['Plotted 38841 homes within the Exeter area. First 5 homes: {'osid\\': '\\b9a49c80-fecf-4308-a98d-cf90ea4ea2e4\\', '\\description\\': '\\End-Of-Terrace House\\', '\\geometry_area_m2\\': 99.338}, {'osid\\': '\\9dfb4b41-f53d-44c0-8a7a-5f7884375677\\', '\\description\\': '\\Mid-Terrace House\\', '\\geometry_area_m2\\': 78.565}, {'osid\\': '\\aea058f2-6a39-4829-ae6b-85c6863a250e\\', '\\description\\': '\\Mid-Terrace House\\', '\\geometry_area_m2\\': 66.91}, {'osid\\': '\\a578be0a-7fff-4fbb-8ace-2cf755bbb53c\\', '\\description\\': '\\End-Of-Terrace House\\', '\\geometry_area_m2\\': 51.92}, {'osid\\': '\\c86c6a23-5ced-4dae-be17-94f33e02c66d\\', '\\description\\': '\\Mid-Terrace House\\', '\\geometry_area_m2\\': 51.91}]", 'homes_within_exeter_map', 'Folium map showing homes within the Exeter area polygon, with Exeter boundary and homes highlighted.', 'homes_within_exeter_map.html']

host_agent

I have found and mapped 38,841 homes within the Exeter area. These homes are residential buildings, each with exactly one address. The map highlights all the homes within the Exeter boundary.

If you would like to see specific details, filter by type, or focus on a particular part of Exeter, please let me know!